

CHE 330 Exam 2 Study Guide

The second CHE 330 exam will be a 50 minute open-book, open-note exam covering the application of statistics and safety to engineering problems. In order to do well on this exam, you should be able to do the following:

Hypothesis Testing (Z-test, t-test, ANOVA)

1. Construct research questions specifying one-tail or two-tail hypothesis tests.
2. Determine whether Z-test, t-test or ANOVA is appropriate for hypothesis testing of given population and/or sample data.
3. Specify appropriate null and alternative hypothesis for hypothesis testing.
4. Determine significance level from a given confidence level (or vice-versa).
5. Calculate Z-score, t-score or F-score from a given set of population and/or sample data.
6. Complete hypothesis testing using Z-statistics, t-statistics or F-statistics and the standard Z-, t-, and F-tables to determine whether to accept or reject the null hypothesis at a given significance/confidence level.

Design of Experiments

1. Specify the number of trials necessary to complete a full factorial experimental design with X factors and Y levels.
2. Evaluate experimental data to determine the mean response, effects of factors, and effects of factor interactions for a 2-level, 3-factor experimental design scheme.
3. Determine which factors and factor interactions contribute most to system response based on experimental data.
4. Construct a model equation predicting system response for a given set of experimental data.
5. Calculate predicted system responses based on a model equation and given values of factors.

Least-Squares Regression

1. Determine a regression model for a given set of linear or non-linear (X, Y) data points.
2. Apply linearization transforms to non-linear trends to determine regression models.
3. Apply regression models to determine a predicted Y-value for a given X-value (or vice-versa).
4. Determine R^2 values signifying regression model fit for a given set of (X, Y) data points.
5. Determine confidence intervals for the slope of a regression model.

Engineering Safety

1. Describe safety incidents evaluated by chosen case studies and their resulting catastrophic effect(s).
2. Identify key factors/causes of safety incidents addressed by case studies.
3. Identify safety precautions that could have avoided safety incidents addressed by case studies.